

HW06 API Testing Report - EShop

1. Thông tin bài làm

Mục	Giá trị
Bài tập	HW06 - API Testing
SUT	EShop
Base URL	http://localhost:3000
Hình thức kiểm thử	Blackbox API testing
Công cụ chính	Postman, Newman, GitHub Actions, Codex
Student ID	23127475
Video demo	https://youtu.be/Aue36T6uS5M
Public GitHub repository	https://github.com/lmchkh/CS423-CSC15003-Testing-N08/tree/hw06/23127475

Nguyên tắc: mọi test case và expected behavior trong report này được thiết kế từ [README.md](#), [api_specification.md](#), API responses quan sát được, và evidence thực thi. Không dùng source code backend/frontend để suy ra expected behavior.

2. API selection

Pool	FR	API chính	API phụ dùng setup/verify/cleanup	Lý do chọn
Pool A	FR-03 Forgot password/reset password	POST /api/reset-password	POST /api/forgot-password	Có nhiều parameter (email , resetToken , newPassword), cover password complexity, OTP lifecycle, SEC-07, schema/error cases.
Pool B	FR-09 Discount coupons	POST /api/apply-coupon	Login user, checkout/order setup nếu cần	Cover C1-C5, code , total_amount , user_id , discount calculation, auth/IDOR risk, schema discount_amount/final_amount .
Pool C	FR-17 Coupon management CRUD	POST /api/admin/coupons	GET /api/coupons , DELETE /api/admin/coupons/:id , admin/user login	Có nhiều input để domain partition và cover admin RBAC SEC-02/SEC-03.

Ghi chú: các FR được phân không chứa FR-10 order state machine, nên state/workflow coverage sẽ dùng lifecycle phù hợp với API đã chọn: OTP lifecycle, coupon conditions/usage, và coupon CRUD lifecycle.

2.1 Artifact layout

Path	Vai trò	Ghi chú
test-cases/hw06-api/	Test case design chi tiết	Mỗi final test case có một file Markdown riêng, dùng template TEMPLATE-HW06-API-TEST-CASE.md .
reports/hw06-test-cases.md	Master test case table	Tổng hợp từng TC ID, file path, audit label, human extension, execution result, evidence.

Path	Vai trò	Ghi chú
postman/	Input artifacts để execute test	Chứa Postman collection, environment và data files. Đặt ở root để CLI/CI path ngắn và dễ chạy.
reports/newman/	Output evidence từ execution	Chứa Newman HTML/JSON reports, nên đặt dưới reports/ cùng các evidence/report khác.
reports/bug-reports/	Bug report Markdown	Mỗi bug report bám template .github/ISSUE_TEMPLATE/bug-report-template.md .

3. API 1 - FR-03 POST /api/reset-password

3.1 Generate with AI

Trạng thái: Đã thực hiện ở Phase 02; raw output đã được human audit ở Phase 03.

Artifact	Link/ghi chú
Test case files	test-cases/hw06-api/fr03-reset-password/
Prompt/output trace	reports/ai-audit-report.md Entry #7; prompt framework dựa trên skills/ và reports/hw06-execution-plan.md
Raw AI test cases	reports/ai-generated/fr03-reset-password-raw-test-cases.md
AI audit entry	reports/ai-audit-report.md Entry #7

Raw generation summary:

Group	Count	Ghi chú
Domain	22	Cover email , resetToken , newPassword partitions
Security	8	Cover SEC-05, SEC-07, sensitive data exposure
Workflow	8	Cover forgot-password setup, token binding/reuse, postcondition login
Schema	8	Cover JSON content type, message/error shape, no sensitive fields, response time
Total	46	Đã audit ở Phase 03; xem reports/hw06-test-cases.md section 3

3.2 Human audit

Trạng thái: Đã thực hiện ở Phase 03. Chi tiết audit: [reports/hw06-test-cases.md](#) section 3.

Tổng AI cases	VALID	INVALID	INCOMPLETE	Corrected final cases
46	39	1	6	44

3.3 Human extension

Trạng thái: Đã thực hiện ở Phase 03. Đã thêm 6 human-authored cases, final FR-03 suite có 50 cases.

TC ID	Missed case	Expected result	Vì sao AI bỏ sót
TC-FR03-API-DOM-023	newPassword empty string	400, có message/error , không 5xx	AI thường gộp empty password với too-short password
TC-FR03-API-DOM-	Valid password đúng boundary 8 ký tự	200, có message , không trả password/token	AI có case too short và valid dài hơn nhưng bỏ sót exact lower boundary

TC ID	Missed case	Expected result	Vì sao AI bỏ sót
024			
TC-FR03-API-DOM-025	resetToken có whitespace đầu/cuối	400, có message/error, không 5xx	AI kiểm tra whitespace email nhưng bỏ sót whitespace token
TC-FR03-API-WF-007	Failed reset do weak password không đổi password hiện tại	400 then old password login 200	AI tập trung token consumption, bỏ sót postcondition dữ liệu password
TC-FR03-API-SCH-009	Malformed JSON body	400, không leak stack trace	AI chủ yếu sinh JSON syntactically valid
TC-FR03-API-SCH-010	Invalid Content-Type: text/plain	400 hoặc 415, không 5xx	AI kiểm tra success Content-Type nhưng bỏ sót request content-type negative

3.4 Execution with Postman/Newman

Trạng thái: Đã thực hiện ở Phase 04 trên backend http://localhost:3000.

Artifact	Path/link
Collection	postman/hw06-fr03-reset-password.postman_collection.json
Environment	postman/hw06-local.postman_environment.json
Case data/traceability file	postman/data/hw06-fr03-reset-password.data.json
Newman HTML report	reports/newman/hw06-fr03-reset-password.html
Newman JSON report	reports/newman/hw06-fr03-reset-password.json
Newman CLI output	reports/newman/hw06-fr03-reset-password-cli.txt
X-Student-Id evidence	50/50 request item có static header X-Student-Id: {{studentId}}; collection-level pre-request script vẫn upsert header X-Student-Id=23127475; Newman CLI output logs [HW06] ... X-Student-Id=23127475 for executed requests.

Execution summary:

Metric	Value
Final test cases	50
Executed test cases	50
Passed test cases	36
Failed test cases	14
Newman requests	136 executed, 0 failed
Newman assertions	218 executed, 20 failed
Confirmed bugs	3

Newman exit code là 1 vì assertion failures từ SUT behavior. Không có request failure hoặc script failure trong run chính.

3.5 Bugs found

Trạng thái: Đã xác nhận 3 bug qua Newman evidence và đã tạo GitHub Issue sau khi sinh viên approve.

Bug ID	Related TC	Git Hub Issue	Evidence
BUG-FR03-API-001	TC-FR03-API-DOM-015 đến DOM-023, SEC-004, WF-006, WF-007	#268	reports/bug-reports/BUG-FR03-API-001.md, Newman HTML/JSON/CLI
BUG-FR03-API-002	TC-FR03-API-SEC-008	#269	reports/bug-reports/BUG-FR03-API-002.md, Newman HTML/JSON/CLI
BUG-FR03-API-003	TC-FR03-API-SCH-010	#270	reports/bug-reports/BUG-FR03-API-003.md, Newman HTML/JSON/CLI

4. API 2 - FR-09 POST /api/apply-coupon

4.1 Generate with AI

Trạng thái: Đã thực hiện ở Phase 05; raw output đã được human audit ở Phase 06.

Artifact	Link/ghi chú
Test case files	test-cases/hw06-api/fr09-apply-coupon/
Prompt/output trace	reports/ai-audit-report.md Entry #9; prompt framework dựa trên skills/ và reports/hw06-execution-plan.md
Raw AI test cases	reports/ai-generated/fr09-apply-coupon-raw-test-cases.md
AI audit entry	reports/ai-audit-report.md Entry #9

Raw generation summary:

Group	Count	Ghi chú
Domain	20	Cover code, total_amount, user_id, sample coupons, min-order boundary, expired/unknown code
Security	10	Cover C4/SEC-02 auth, IDOR/body user_id mismatch, SQLi/XSS, sensitive-field leak
Workflow	8	Cover C5 per-user usage limit, repeated use, failed apply should not consume usage
Schema	8	Cover discount_amount, final_amount, JSON content type, calculation, error shape, response time
Total	46	Sẽ audit ở Phase 06; có assumptions cần xác nhận bằng blackbox observation

4.2 Human audit

Trạng thái: Đã thực hiện ở Phase 06. Chi tiết audit: reports/hw06-test-cases.md section 4.

Tổng AI cases	VALID	INVALID	INCOMPLETE	Corrected final cases
46	39	0	7	45

4.3 Human extension

Trạng thái: Đã thực hiện ở Phase 06. Đã thêm 6 human-authored cases, final FR-09 suite có 51 cases.

TC ID	Missed case	Expected result	Vì sao AI bỏ sót
TC-FR09-API-DOM-021	Valid coupon code có whitespace đầu/cuối	4xx, có message/error nếu JSON, không có success discount fields	AI có whitespace-only và lowercase nhưng bỏ sót valid code kèm whitespace
TC-FR09-API-DOM-022	user_id là null	400, có message/error	AI có missing user_id và IDOR nhưng bỏ sót null partition
TC-FR09-API-DOM-023	user_id là string numeric	400, có message/error	AI có SQLi string user_id nhưng bỏ sót string numeric trông hợp lệ
TC-FR09-API-SEC-010	Token không có Bearer prefix	401, có message/error	AI có missing/malformed Bearer nhưng bỏ sót token không có Bearer prefix
TC-FR09-API-SCH-009	Content-Type: text/plain với JSON-looking body	400 hoặc 415, không success, không 5xx	AI có malformed JSON nhưng bỏ sót unsupported request Content-Type
TC-FR09-API-SCH-010	Success response không expose internal coupon config fields	200, không trả id, type, discount_value, min_order_amount, expired_at, max_uses_per_user, is_active	AI kiểm tra token/password leak nhưng bỏ sót internal coupon configuration leak

4.4 Execution with Postman/Newman

Trạng thái: Đã thực hiện ở Phase 07 trên backend <http://localhost:3000>.

Artifact	Path/link
Collection	postman/hw06-fr09-apply-coupon.postman_collection.json
Environment	postman/hw06-local.postman_environment.json
Case data/traceability file	postman/data/hw06-fr09-apply-coupon.data.json
Newman HTML report	reports/newman/hw06-fr09-apply-coupon.html
Newman JSON report	reports/newman/hw06-fr09-apply-coupon.json
Newman CLI output	reports/newman/hw06-fr09-apply-coupon-cli.txt
X-Student-Id evidence	51/51 request item có static header X-Student-Id: {{studentId}} ; collection-level pre-request script upsert header và Newman CLI output logs [HW06] ... X-Student-Id=23127475 .

Execution summary:

Metric	Value
Final test cases	51
Executed test cases	51

Metric	Value
Passed test cases	27
Failed test cases	24
Newman requests	165 executed, 0 failed
Newman assertions	241 executed, 35 failed
Confirmed bugs	8

Newman exit code là 1 vì assertion failures từ SUT behavior. Không có request failure, prerequest script failure hoặc test script failure trong run chính.

4.5 Bugs found

Trạng thái: Đã xác nhận 8 bug qua Newman evidence và đã tạo GitHub Issues #271-#278 sau khi sinh viên review.

Bug ID	Related TC	GitHub Issue	Evidence
BUG-FR09-API-001	TC-FR09-API-DOM-001, SEC-008, WF-005 đến WF-008, SCH-003	#271	reports/bug-reports/BUG-FR09-API-001.md, Newman HTML/JSON/CLI
BUG-FR09-API-002	TC-FR09-API-DOM-002 đến DOM-004	#272	reports/bug-reports/BUG-FR09-API-002.md, Newman HTML/JSON/CLI
BUG-FR09-API-003	TC-FR09-API-SEC-001 đến SEC-003, SEC-010	#273	reports/bug-reports/BUG-FR09-API-003.md, Newman HTML/JSON/CLI
BUG-FR09-API-004	TC-FR09-API-SEC-004, WF-008	#274	reports/bug-reports/BUG-FR09-API-004.md, Newman HTML/JSON/CLI
BUG-FR09-API-005	TC-FR09-API-DOM-019, DOM-020, DOM-022, DOM-023, SEC-006	#275	reports/bug-reports/BUG-FR09-API-005.md, Newman HTML/JSON/CLI
BUG-FR09-API-006	TC-FR09-API-WF-001, WF-004	#276	reports/bug-reports/BUG-FR09-API-006.md, Newman HTML/JSON/CLI
BUG-FR09-API-007	TC-FR09-API-SCH-009	#277	reports/bug-reports/BUG-FR09-API-007.md, Newman HTML/JSON/CLI
BUG-FR09-API-008	TC-FR09-API-SCH-010	#278	reports/bug-reports/BUG-FR09-API-008.md, Newman HTML/JSON/CLI

5. API 3 - FR-17 POST /api/admin/coupons

5.1 Generate with AI

Trạng thái: Đã thực hiện ở Phase 08; raw output đã được human audit ở Phase 09.

Artifact	Link/ghi chú
Test case files	test-cases/hw06-api/fr17-admin-coupons/
Prompt/output trace	reports/ai-audit-report.md Entry #12; prompt framework dựa trên skills/ và reports/hw06-execution-plan.md
Raw AI test cases	reports/ai-generated/fr17-admin-coupons-raw-test-cases.md
AI audit entry	reports/ai-audit-report.md Entry #12

Raw generation summary:

Group	Count	Ghi chú
Domain	30	Cover <code>code</code> , <code>type</code> , <code>discount_value</code> , <code>min_order_amount</code> , <code>expired_at</code> , <code>max_uses_per_user</code> , uniqueness, valid percent/fixed create
Security	10	Cover SEC-02/SEC-03 admin RBAC, missing/malformed/invalid token, user token, SQLi/XSS, role escalation, sensitive leak
Workflow	7	Cover create -> list verify -> duplicate reject -> cleanup delete, failed create no partial record
Schema	7	Cover JSON content type, success/error shape, malformed JSON, unsupported Content-Type, method contract, response time
Total	54	Đã audit ở Phase 09; assumptions được xử lý bằng label <code>VALID/INVALID/INCOMPLETE</code> và final cases

5.2 Human audit

Trạng thái: Đã thực hiện ở Phase 09. Chi tiết audit: `reports/hw06-test-cases.md` section 5.

Tổng AI cases	VALID	INVALID	INCOMPLETE	Corrected final cases
54	40	1	13	50

5.3 Human extension

Trạng thái: Đã thực hiện ở Phase 09. Đã thêm 6 human-authored cases, final FR-17 suite có 56 cases.

TC ID	Missed case	Expected result	Vì sao AI bỏ sót
TC-FR17-API-DOM-028	<code>expired_at</code> là null	400; message hoặc error; không tạo coupon	AI có missing/malformed/past date nhưng bỏ sót null date.
TC-FR17-API-DOM-029	<code>discount_value</code> là null	400; message hoặc error; không tạo coupon	AI có missing/zero/negative/string nhưng bỏ sót null discount_value.
TC-FR17-API-DOM-030	<code>min_order_amount</code> là null	400; message hoặc error; không tạo coupon	AI có missing/negative/string nhưng bỏ sót null min_order_amount.
TC-FR17-API-DOM-031	<code>max_uses_per_user</code> là null	400; message hoặc error; không tạo coupon	AI có missing/zero/string nhưng bỏ sót null max_uses_per_user.
TC-FR17-API-DOM-032	<code>type</code> là empty string	400; message hoặc error; không tạo coupon	AI có missing/null/unsupported/wrong-case type nhưng bỏ sót empty string.
TC-FR17-API-WF-008	Deleted coupon code có thể được tạo lại sau cleanup	200/201 then 200/204 then 200/201; Second create succeeds; final cleanup bằng DELETE	AI có delete/list verify nhưng bỏ sót recreate-after-delete lifecycle.

5.4 Execution with Postman/Newman

Trạng thái: Đã thực hiện ở Phase 10 bằng Newman local với backend `http://localhost:3000`.

Artifact	Path/link
Collection	postman/hw06-fr17-admin-coupons.postman_collection.json
Environment	postman/hw06-local.postman_environment.json
Iteration data	postman/data/hw06-fr17-admin-coupons.data.json
Newman HTML report	reports/newman/hw06-fr17-admin-coupons.html
Newman JSON report	reports/newman/hw06-fr17-admin-coupons.json
Newman CLI output	reports/newman/hw06-fr17-admin-coupons-cli.txt
X-Student-Id evidence	Newman CLI/HTML include collection pre-request logs with X-Student-Id=23127475

Execution result:

Final cases	Executed	Passed	Failed	Bugs confirmed
56	56	18	38	6

Triage note: initial Newman run exposed one test-oracle issue for TC-FR17-API-SEC-005 (code SQLi payload). Because FR-17 does not define a code charset, the final assertion was corrected to require no 5xx/no SQL leak and cleanup if the payload is stored as a literal. The final run above maps all remaining 38 failed final cases to SUT bugs in section 5.5.

5.5 Bugs found

Trạng thái: Đã tạo bug report Markdown ở Phase 10 và đã tạo GitHub Issues #279-#284 sau khi sinh viên review.

Bug ID	Related TC	GitHub Issue	Evidence
BUG-FR17-API-001	Invalid required/range/type partitions	#282	reports/bug-reports/BUG-FR17-API-001.md, Newman HTML/JSON/CLI
BUG-FR17-API-002	Duplicate coupon code returns 500	#280	reports/bug-reports/BUG-FR17-API-002.md, Newman HTML/JSON/CLI
BUG-FR17-API-003	User token can create admin coupons	#283	reports/bug-reports/BUG-FR17-API-003.md, Newman HTML/JSON/CLI
BUG-FR17-API-004	Invalid JWT returns 403 instead of 401	#279	reports/bug-reports/BUG-FR17-API-004.md, Newman HTML/JSON/CLI
BUG-FR17-API-005	Malformed JSON leaks HTML stack trace	#281	reports/bug-reports/BUG-FR17-API-005.md, Newman HTML/JSON/CLI
BUG-FR17-API-006	Unsupported text/plain crashes endpoint	#284	reports/bug-reports/BUG-FR17-API-006.md, Newman HTML/JSON/CLI

6. Postman features used

Trạng thái: Đã dùng cho FR-03, FR-09 và FR-17.

Feature	Đã dùng?	Evidence/ghi chú
Collections	Có	postman/hw06-fr03-reset-password.postman_collection.json, postman/hw06-fr09-apply-coupon.postman_collection.json, postman/hw06-fr17-admin-coupons.postman_collection.json

Feature	Đã dùng?	Evidence/ghi chú
Environments	Có	<code>postman/hw06-local.postman_environment.json</code>
Environment variables	Có	<code>baseUrl</code> , <code>studentId</code> , generated email/token variables trong environment/runtime
Collection variables	Có	<code>collectionRunId</code> và helper values dùng trong pre-request setup
Pre-request scripts	Có	Setup blackbox bằng API: register/login/reset flows, admin/user token setup, coupon helper setup; upsert <code>X-Student-Id</code>
Test scripts/assertions	Có	Status, schema/message/error shape, sensitive-field leak, token reuse, password postcondition, coupon formula, auth/IDOR/RBAC, usage-limit, response time, coupon cleanup/list verification
Data-driven runs	Một phần	Data files trong <code>postman/data/</code> dùng làm case map/traceability; collections chạy stateful per item để tránh token/state nhiễu nhau
Newman CLI	Có	<code>reports/newman/hw06-fr03-reset-password-cli.txt</code> , <code>reports/newman/hw06-fr09-apply-coupon-cli.txt</code> , <code>reports/newman/hw06-fr17-admin-coupons-cli.txt</code>
HTML/JSON reporters	Có	FR-03, FR-09 và FR-17 đều có HTML/JSON reports trong <code>reports/newman/</code>
GitHub Actions CI	Có	<code>.github/workflows/newman-api-test.yml</code> ; pass/fail evidence ở section 7
Workspaces	Chưa	Chưa có
Monitors/mock servers	Chưa	Optional

7. CI/CD report

Trạng thái: Phase 11 đã thêm GitHub Actions workflow cho passing Newman smoke run trên đúng branch `hw06/23127475`. Phase 12 đổi một assertion status có chủ đích để tạo failing CI evidence. Full FR-03/FR-09/FR-17 bug-finding suites không dùng làm pass gate vì chúng intentionally fail để ghi nhận bugs thật; workflow pass dùng smoke collection để chứng minh checkout, backend startup, Newman CLI, reporters, artifacts và `X-Student-Id`.

Workflow path: `.github/workflows/newman-api-test.yml`

Branch scope: chỉ chạy khi push lên `hw06/23127475`; job cũng có guard `github.ref == 'refs/heads/hw06/23127475'`.

Backend startup command:

```
cd backend
node server.js
```

Newman pass command:

```
newman run postman/hw06-ci-smoke.postman_collection.json \
  --environment postman/hw06-local.postman_environment.json \
  --reporters cli,html,extra,json \
```

```
--reporter-htmlextra-export reports/newman/hw06-ci-smoke.html \
--reporter-json-export reports/newman/hw06-ci-smoke.json
```

Run type	Commit	Workflow URL	Result	Screenshot/evidence	Notes
Passing run	93e35e6	https://github.com/lmchkh/CS423-CSC15003-Testing-N08/actions/runs/32574378458	Passed	Screenshot: reports/screenshots/ci-pass.png ; local evidence: reports/newman/hw06-ci-smoke.html , reports/newman/hw06-ci-smoke.json , reports/newman/hw06-ci-smoke-cli.txt ; uploaded artifact hw06-newman-smoke-reports	Phase 11
Intentional failing run	c8dfd58	https://github.com/lmchkh/CS423-CSC15003-Testing-N08/actions/runs/32575114869	Failed	Screenshot: reports/screenshots/ci-fail.png ; intentional change: in postman/hw06-ci-smoke.postman_collection.json , request CI-SMOKE-001 expects HTTP 999 instead of 200; local evidence: reports/newman/hw06-ci-smoke-intentional-fail.html , reports/newman/hw06-ci-smoke-intentional-fail.json , reports/newman/hw06-ci-smoke-intentional-fail-cli.txt	Phase 12
Restored passing run	ecb0b01	https://github.com/lmchkh/CS423-CSC15003-Testing-N08/actions/runs/32575713163	Passed	Restored change: CI-SMOKE-001 expects HTTP 200 again; local evidence: reports/newman/hw06-ci-smoke-restored-pass.html , reports/newman/hw06-ci-smoke-restored-pass.json , reports/newman/hw06-ci-smoke-restored-pass-cli.txt	Phase 13

8. Bug reports

Bug report rule: khi phát hiện bug thật, tạo Markdown bug report trong [reports/bug-reports](#) theo template [.github/ISSUE_TEMPLATE/bug-report-template.md](#). GitHub Issue chỉ tạo sau khi sinh viên review bug report và yêu cầu tạo issue; sau đó cập nhật URL thật vào report.

Bug ID	Endpoint	Requirement	Severity/Priority	GitHub Issue	Evidence
BUG-FR03-API-001	POST /api/reset-password	FR-03, FR-01, SEC-07	Major/P1	#268	reports/bug-reports/BUG-FR03-API-001.md ; reports/newman/hw06-fr03-reset-password.html

Bug ID	Endpoint	Requirement	Severity/Priority	GitHub Issue	Evidence
BUG-FR03-API-002	POST /api/forgot-password setup for reset flow	FR-03, SEC-07	Major/P1	#269	reports/bug-reports/BUG-FR03-API-002.md; reports/newman/hw06-fr03-reset-password.html
BUG-FR03-API-003	POST /api/reset-password	FR-03, SEC-05	Major/P2	#270	reports/bug-reports/BUG-FR03-API-003.md; reports/newman/hw06-fr03-reset-password.html
BUG-FR09-API-001	POST /api/apply-coupon	FR-09 formula	Major/P1	#271	reports/bug-reports/BUG-FR09-API-001.md; reports/newman/hw06-fr09-apply-coupon.html
BUG-FR09-API-002	POST /api/apply-coupon	FR-09 C3	Major/P1	#272	reports/bug-reports/BUG-FR09-API-002.md; reports/newman/hw06-fr09-apply-coupon.html
BUG-FR09-API-003	POST /api/apply-coupon	FR-09 C4, SEC-02	Critical/P1	#273	reports/bug-reports/BUG-FR09-API-003.md; reports/newman/hw06-fr09-apply-coupon.html
BUG-FR09-API-004	POST /api/apply-coupon	FR-09 C4/C5, SEC-02	Critical/P1	#274	reports/bug-reports/BUG-FR09-API-004.md; reports/newman/hw06-fr09-apply-coupon.html
BUG-FR09-API-005	POST /api/apply-coupon	FR-09 C3-C5, SEC-05	Major/P1	#275	reports/bug-reports/BUG-FR09-API-005.md; reports/newman/hw06-fr09-apply-coupon.html
BUG-FR09-API-006	POST /api/apply-coupon	FR-09 C5	Major/P1	#276	reports/bug-reports/BUG-FR09-API-006.md; reports/newman/hw06-

Bug ID	Endpoint	Requirement	Severity/Priority	GitHub Issue	Evidence
					fr09-apply-coupon.html
BUG-FR09-API-007	POST /api/apply-coupon	FR-09, SEC-05	Major/P2	#277	reports/bug-reports/BUG-FR09-API-007.md; reports/newman/hw06-fr09-apply-coupon.html
BUG-FR09-API-008	POST /api/apply-coupon	API schema	Major/P2	#278	reports/bug-reports/BUG-FR09-API-008.md; reports/newman/hw06-fr09-apply-coupon.html
BUG-FR17-API-001	POST /api/admin/coupons	FR-17 required/range/type validation	Critical/P0	#282	reports/bug-reports/BUG-FR17-API-001.md; reports/newman/hw06-fr17-admin-coupons.html
BUG-FR17-API-002	POST /api/admin/coupons	FR-17 unique code rule	Major/P1	#280	reports/bug-reports/BUG-FR17-API-002.md; reports/newman/hw06-fr17-admin-coupons.html
BUG-FR17-API-003	POST /api/admin/coupons	FR-12, FR-17, SEC-02, SEC-03	Critical/P0	#283	reports/bug-reports/BUG-FR17-API-003.md; reports/newman/hw06-fr17-admin-coupons.html
BUG-FR17-API-004	POST /api/admin/coupons	FR-12, SEC-02	Minor/P2	#279	reports/bug-reports/BUG-FR17-API-004.md; reports/newman/hw06-fr17-admin-coupons.html
BUG-FR17-API-005	POST /api/admin/coupons	FR-17 schema validation, SEC-05	Major/P1	#281	reports/bug-reports/BUG-FR17-API-005.md; reports/newman/hw06-fr17-admin-coupons.html
BUG-FR17-	POST /api/admin/coupons	FR-17 content-type/schema validation	Major/P1	#284	reports/bug-reports/BUG-FR17-API-006.md;

Bug ID	Endpoint	Requirement	Severity/Priority	GitHub Issue	Evidence
API-006					reports/newman/hw06-fr17-admin-coupons.html

9. AI-driven test generator design

Trạng thái: Đã finalize phần design, pseudocode và diagram ở Phase 15. Diagram được sinh viên tự vẽ bằng draw.io và export thành PNG.

Artifact	Path/link	Status
Design document	ai-test-generator-design.md	Hoàn tất
Pseudocode	ai-test-generator-design.md section 3	Hoàn tất
Self-drawn diagram	AI-Driven-API-Test-Generator.drawio.png	Hoàn tất

10. Video demo

Link video demo: <https://youtu.be/Aue36T6uS5M>

11. AI critique

File: [./ai-critique.md](#)

12. AI Audit Report

File: [./ai-audit-report.md](#)

13. Test summary

Trạng thái: Đã có số liệu execution cho FR-03, FR-09 và FR-17.

API	Generated cases	Human added cases	Final cases	Executed	Passed	Failed	Bugs
FR-03 POST /api/reset-password	46	6	50	50	36	14	3
FR-09 POST /api/apply-coupon	46	6	51	51	27	24	8
FR-17 POST /api/admin/coupons	54	6	56	56	18	38	6
Total	146	18	157	157	81	76	17

14. Self-assessment

No.	Criteria	Grade	Self-assessed grade
1	API 1 - full pipeline	30	30
2	API 2 - full pipeline	30	30
3	API 3 - full pipeline	30	30
4	Agent Skills / AI-driven generator	10	10

No.	Criteria	Grade	Self-assessed grade
	Total	100	100

15. Submission checklist

Deliverable	Status	Path/link
Main report Markdown	Có	reports/main-report.md
Main report PDF	Có	reports/main-report.pdf
Public GitHub repository link	Có	https://github.com/lmchkh/CS423-CSC15003-Testing-N08/tree/hw06/23127475
Postman collections	Có FR-03, FR-09, FR-17	postman/hw06-fr03-reset-password.postman_collection.json , postman/hw06-fr09-apply-coupon.postman_collection.json , postman/hw06-fr17-admin-coupons.postman_collection.json
Postman environment/data files	Có FR-03, FR-09, FR-17	postman/hw06-local.postman_environment.json , postman/data/hw06-fr03-reset-password.data.json , postman/data/hw06-fr09-apply-coupon.data.json , postman/data/hw06-fr17-admin-coupons.data.json
Newman HTML reports	Có FR-03, FR-09, FR-17	reports/newman/hw06-fr03-reset-password.html , reports/newman/hw06-fr09-apply-coupon.html , reports/newman/hw06-fr17-admin-coupons.html
Postman feature list	Có FR-03, FR-09	Section 6
CI/CD report	Có	Section 7 có Phase 11 pass URL, Phase 12 intentional fail URL và Phase 13 restored pass URL
Excel/test case table	Có	reports/hw06-test-cases.md , test-cases/hw06-api/ , reports/HW06-API-Testing.xlsx
AI test-generator diagram/pseudocode	Có	ai-test-generator-design.md , AI-Driven-API-Test-Generator.drawio.png
Bug reports/GitHub Issues	Có 3 bug reports FR-03 đã tạo issue; có 8 bug reports FR-09 đã tạo issue; có 6 bug reports FR-17 đã tạo issue	reports/bug-reports/ , #268-#284
Video demo	Có	Section 10
AI Critique MD	Có	reports/ai-critique.md
AI Critique PDF	Có	reports/ai-critique.pdf
AI Audit Report MD	Có	reports/ai-audit-report.md
AI Audit Report PDF	Có	reports/ai-audit-report.pdf
Git commit log	Có	git-commit-log.md
README self-assessment	Có	reports/README.md